

CoRoT-2b

R-0074

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_250504 · created 2026-07-15T19:40:08+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460799.9274 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.037
Transit depth (Rp/Rs)^2	2.77 +/- 1.23 [%]
Orbital Inclination (inc)	85.15 +/- 1.61
Airmass coefficient 1 (a1)	0.957 +/- 0.012
Airmass coefficient 2 (a2)	0.03 +/- 0.11
Scatter in the residuals of the lightcurve fit is	1.27 %
Best Comparison Star	#2 - [553, 422]
Optimal Aperture	3.65
Optimal Annulus	10.99
Transit Duration (day)	0.0453 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.166 $\pm$ 0.037 vs 0.1667 (deviation 0.01 σ)
Duration fit vs published	0.0453 d vs 0.095 d (deviation 5.75 σ)
Mid-time O-C	10.4 min
Depth SNR	3.0
Residual scatter	1.27 %

Light curve

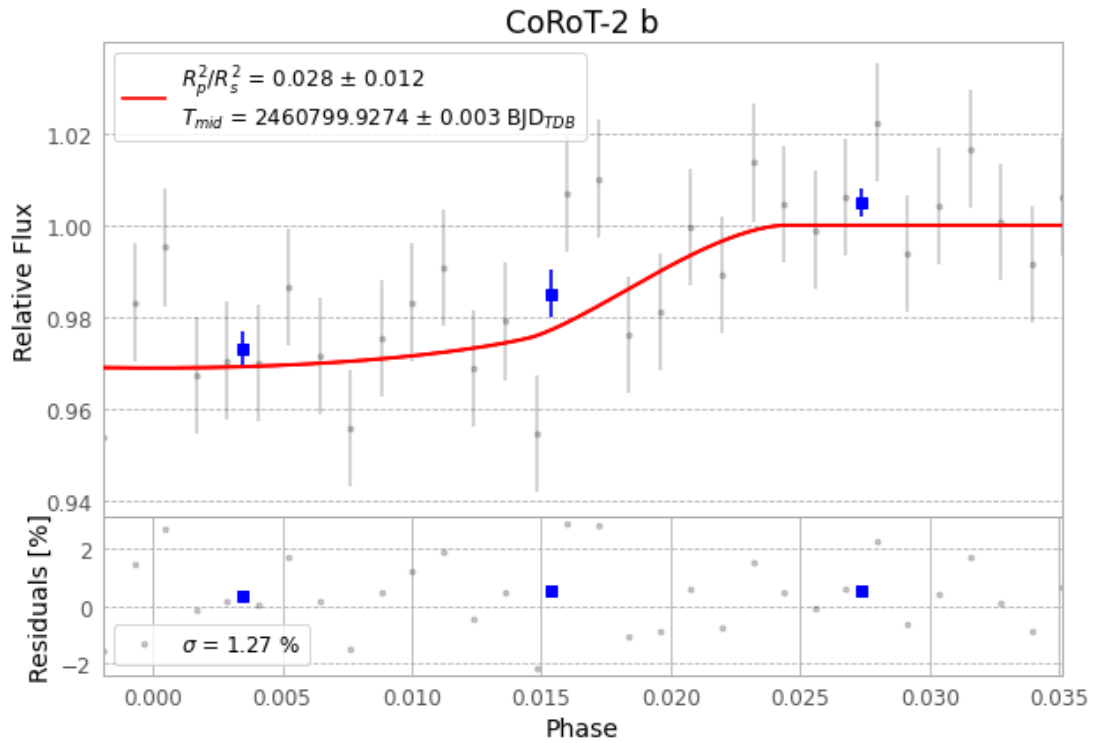


Figure 1 — FinalLightCurve_CoRoT-2 b_2025-05-04.png (SHA-256 5d6dba49f744b981...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [346, 218], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dddcf847fb1a5296...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

32 FITS frames (21 MB), CoRoT-2250504100615.FITS → CoRoT-2250504113915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-250504.log (48114414ad3b...), FinalLightCurve_CoRoT-2 b_2025-05-04.png (5d6dba49f744...), FinalLightCurve_CoRoT-2 b_2025-05-04.csv (b504ef541d6e...)

Compute resources

Wall time 175.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 19:40Z.